Two Illustrative Clinical Learning Episodes

รายงานการประเมิน PIS-7 · ตัวอย่างบทสนทนาแพทย์–AI จำนวน 2 ชุด

Both transcripts below were scored against the **production PIS-7 evaluator rubric v1** and the production domain weights. The rubric scores the **physician's** reasoning behaviour only — the AI's turns are read as context, never as credit. The two episodes were written to differ in learning depth, and the rubric separates them cleanly: **48.0** versus **93.0** on the 0–100 composite.

48.0
Example 1 — Diarrhoea cluster
Meaningful-learning tier
5 physician turns

93.0
Example 2 — HIV positivity signal
High-depth episode
6 physician turns

Method · วิธีประเมิน

Rubric. `PIS7_EVALUATOR_PROMPT` (v1, the calibrated prompt) from `rimpl/src/lib/prompts.js` — seven domains, each 0–5, with the v1 calibration rules applied before scoring: *structure is not depth*; templated behaviour is penalised; and a 4–5 in any domain requires a quotable physician utterance that could not have been written without engaging this particular case.

Composite. `compositeScore()` from `backend/src/services/cle-gate.js`: the weighted sum of the seven domain scores × 20, giving a 0–100 scale where 5 across all domains = 100.

Evidence rule. Every 4–5 awarded below is followed by the verbatim physician quote that earned it. Where no such quote exists, the domain is capped at 3 regardless of how competent the exchange reads.

RESULTS

Scores side by side

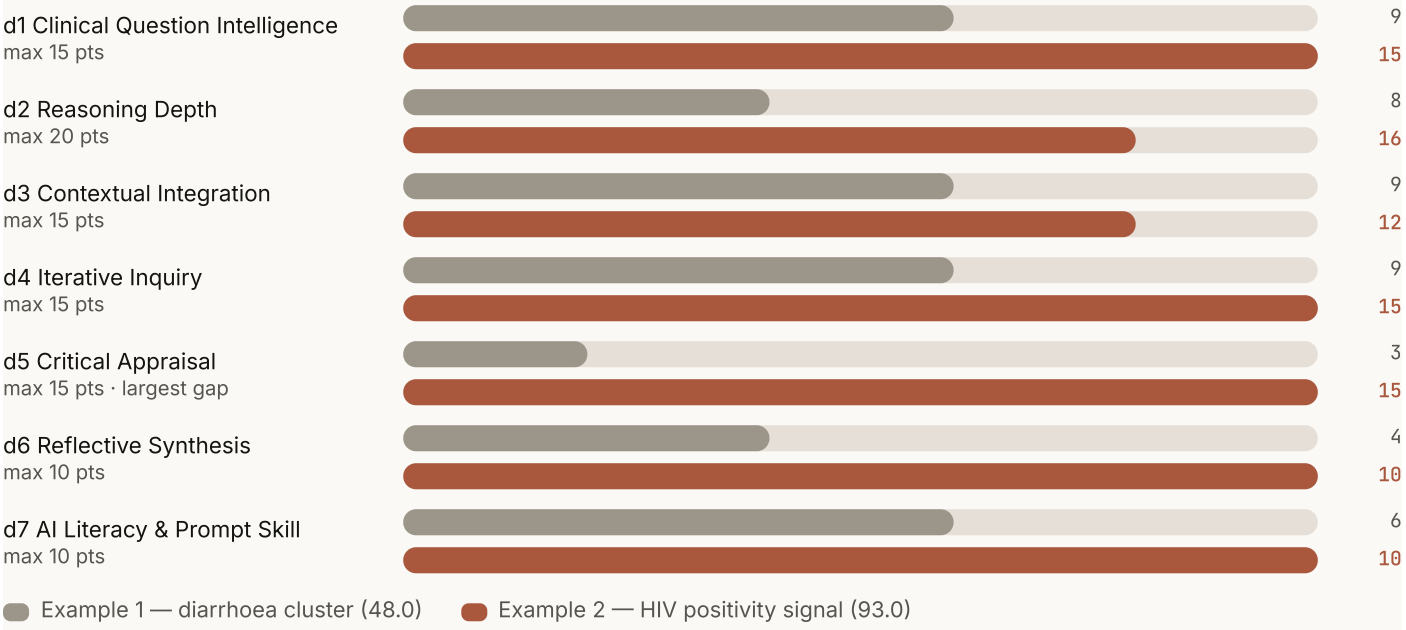
Seven domains, 0–5 each, weighted to a 0–100 composite.

Domain-by-domain comparison

PIS-7 domain	Weight	Ex. 1	Ex. 2	Pts Ex. 1	Pts Ex. 2
d1 · Clinical Question Intelligence	15%	3	5	9.0	15.0
d2 · Reasoning Depth	20%	2	4	8.0	16.0
d3 · Contextual Integration	15%	3	4	9.0	12.0
d4 · Iterative Inquiry	15%	3	5	9.0	15.0
d5 · Critical Appraisal	15%	1	5	3.0	15.0
d6 · Reflective Synthesis	10%	2	5	4.0	10.0
d7 · AI Literacy & Prompt Skill	10%	3	5	6.0	10.0
Composite (0–100)	**100%**	**2.40 avg**	**4.65 avg**	**48.0**	**93.0**

Where the 45-point gap comes from

Weighted points earned out of each domain's maximum. A single domain — **Critical Appraisal** — accounts for 12 of the 45-point difference.



SCORECARD — EXAMPLE 1

Acute watery diarrhoea cluster

Village health post · 5 physician turns · composite **48.0 / 100**

A competent, appropriate consultation that produces the right public-health action — and still scores in the mid-40s, because the rubric measures reasoning behaviour rather than outcome. The physician supplies good local data but never generates a hypothesis of their own, never tests the AI's answer, and closes without naming what changed.

d1 Clinical Question Intelligence

15% · 3

Turns 1 and 3 are anchored to a real setting and expose a genuine decision point — the opening carries case counts, timeframe, population and household clustering rather than a generic ask. But turn 2 is explicitly framed as a general procedural lookup, which the rubric scores ≤2.

> **T3 · earns the 3:** *"The households affected do share a shallow well. Does that change your thinking?"*

> **T2 · caps it at 3:** *"Is there a general approach for deciding whether this counts as an outbreak that needs to be reported?"*

d2 Reasoning Depth

20% · 2

The physician never commits to a ranked hypothesis, never weighs one aetiology against another, and never states what would change their mind. They supply data and accept the AI's framing wholesale. The rubric caps an unranked, unjustified differential at 2 — here there is no differential of the physician's own at all, so 2 is the ceiling and the floor.

d3 Contextual Integration

15% · 3

The shared shallow well is decision-altering context: it redirects the AI toward waterborne pathogens and the physician converts it into safe-water messaging. That is real integration. It stays at 3 because the context arrives once, reactively, and nothing else about the setting — the health post's testing capacity, referral distance, the affected adults' risk profile — is ever allowed to shape the plan.

d4 Iterative Inquiry

15% · 3

Turn 3 genuinely builds on the AI's opening common-source hypothesis by testing it against a specific shared exposure. Turn 2 partly sidesteps into procedure, and turns 4–5 are acknowledgements rather than probes — the inquiry stops narrowing after the third turn.

Acute watery diarrhoea cluster

Domains d5–d7 · composite **48.0 / 100**

d5 Critical Appraisal

15% · 1

Nothing the AI says is questioned, graded, or pushed back on. No evidence is requested, no uncertainty is named, no alternative is raised. The rubric is explicit: accepting everything unexamined is ≤1. This single domain costs the episode 12 of the 15 available points.

> **T4 · pure acceptance:** *"Understood — I'll notify the district health office and advise the households on safe water practices in the meantime."*

d6 Reflective Synthesis

10% · 2

The closing states a next action but names no learning: nothing about the physician's approach is identified as having changed, and the takeaway is not transferable to the next surveillance signal. It scores above the generic "that was useful" floor of 1, but only just.

> **T5 · closing:** *"Good to know. I'll keep monitoring case counts over the next few days."*

d7 AI Literacy & Prompt Skill

10% · 3

Prompts are clear, clinically framed and efficient, and the physician retains ownership of the final action — the decision to notify the district health office is theirs, not the AI's. What is missing is any verification behaviour: no output is checked, no evidence grade is requested, no limitation of the AI's answer is acknowledged.

What would move Example 1 up

Three additions, none of which lengthen the consultation much, would raise the composite from 48.0 to roughly 75 — and they are the same three behaviours the rubric is designed to elicit.

Domain	Missing behaviour	Now	Plausible
d5 Critical Appraisal	Ask for the evidence behind "common-source exposure", or name what would argue against the well — e.g. cases in households without well access.	1	4
d2 Reasoning Depth	Commit to a ranked differential — cholera vs. rotavirus vs. food-borne toxin — with a reason for the ranking and a stated trigger for revising it.	2	4
d6 Reflective Synthesis	Close with the transferable rule, not the next task: what about handling household clusters has changed for the next time one appears.	2	4
Resulting composite			**≈ 75.0**

Example 1 · composite 48.0 = (3×.15 + 2×.20 + 3×.15 + 3×.15 + 1×.15 + 2×.10 + 3×.10) × 20

SCORECARD — EXAMPLE 2

Rising HIV test positivity

Community clinic surveillance · 6 physician turns · composite **93.0 / 100**

This episode does the three things the rubric rewards most and Example 1 never does: it **commits** to a competing hypothesis, it **challenges** the AI and demands graded evidence, and it **closes with a transferable rule** the physician can carry to the next signal. Every 4–5 below is backed by a quotable, case-specific utterance.

d1 Clinical Question Intelligence

15% · 5

Each question is bound to this clinic's actual signal — a quantified rise, over a stated period, against a named comparator — and each exposes a live decision point rather than requesting a recital. Turn 3 goes further and specifies the *form* the answer must take.

> **T3 · earns the 5:** *"Is there a standard epidemiological approach for telling apart a real increase in incidence from a testing artifact? I want an evidence grade on that, not just a general statement."*

d2 Reasoning Depth

20% · 4

A full commit–hold–revise cycle: the physician commits to a specific competing hypothesis at turn 2, holds it against the AI's framing at turn 3, then revises to an evidence-gated position at turn 5 once stratification is on the table. That is genuine hypothesis revision under new information. It stops at 4 rather than 5 because the reasoning is never made explicitly probabilistic — no prior is stated, no threshold is committed to, and the risk of acting on a false signal is never weighed against the cost of delay.

> **T2 · the commitment:** *"I think there may be a genuine increase in transmission in a specific risk group here."*

d3 Contextual Integration

15% · 4

Turn 4 injects programmatic context that could not be guessed from outside this setting — a new outreach programme aimed at migrant workers, plus informal local reports — and it visibly redirects the analysis into stratification by testing channel, which the physician then adopts as their plan. Held at 4 because that integration happens once: the mobility, follow-up and linkage-to-care implications of a migrant-worker cohort never re-enter the reasoning.

> **T4 · earns the 4:** *"This area recently started outreach testing targeting migrant workers, and there have been informal reports of increased risk behavior locally."*

d4 Iterative Inquiry

15% · 5

Every turn engages the specific content of the one before it — turn 2 contests turn 1's framing, turn 3 demands the method behind turn 2's suggestion, turn 4 answers the question that method raises, turn 5 adopts the stratification it produced. No restart, no topic-hop, continuous narrowing from "what could explain this?" to a single named dataset.

Rising HIV test positivity

Domains d5–d7 · composite **93.0 / 100**

d5 Critical Appraisal

15% · 5

The strongest domain in either transcript. The physician directly challenges the AI's framing, explicitly rejects an ungraded answer, and constrains their own conclusion pending evidence. The rubric's instruction not to penalise appraisal for bluntness applies squarely to turn 2.

> **T2 · earns the 5:** *"Are you just attributing this to testing bias?"*

d6 Reflective Synthesis

10% · 5

A textbook 5: the takeaway is in the physician's own words, grounded in this specific case, and stated as a rule that changes how the *next* surveillance signal will be handled — including the condition that triggers it. Nothing about it is generic.

> **T6 · earns the 5:** *"I won't conclude that a rise in detection reflects a real increase in transmission until I've ruled out changes in testing coverage or case-finding strategy, especially when a new outreach program has started around the same time."*

d7 AI Literacy & Prompt Skill

10% · 5

Prompts are precise and specify the required output form; the AI's confidence is treated as something to be tested rather than accepted; and clinical ownership is unambiguous — the physician, not the AI, sets the evidentiary bar that must be cleared before a conclusion is drawn.

> **T5 · ownership:** *"Then I'll request the positivity data stratified by testing channel before concluding this represents a real increase in transmission."*

C1–C6 gate and what it does not see

Before PIS-7 scoring, a CLE passes the six-criterion Tier-1 screen. C1–C3 are mechanical session metrics; C4–C6 are behavioural and classified from the transcript. Applied to these two excerpts:

Crit.	Threshold	Example 1	Example 2
C1	Duration ≥ 600 s (10 min) active engagement	Not determinable — no timestamps	Not determinable — no timestamps
C2	Turn count ≥ 4	PASS 5 turns	PASS 6 turns
C3	Cumulative token count ≥ 10,000	BELOW ≈386 tok	BELOW ≈711 tok
C4	≥ 1 clinical question that is not a factual lookup	PASS shared-well probe	PASS artifact vs. incidence
C5	≥ 2 follow-ups building on prior AI content	MARGINAL 2, one weak	PASS 4 clear
C6	≥ 1 explicit learning summary from the physician	FAIL action, not learning	PASS explicit rule change

READING THE C3 RESULT

Both excerpts fall far below the 10,000-token floor, but this is **not** evidence that either session would fail in production. C3 counts *billed* session tokens — system prompt plus the full conversation re-sent on every turn — which accumulates several times faster than raw transcript length. These are illustrative excerpts, not instrumented sessions; C1 and C3 can only be evaluated from session metadata.

READING THE C6 RESULT

Example 1's closing states a next *action* ("I'll keep monitoring case counts") rather than a *learning*. The C6 classifier looks for an explicit takeaway — "so I've learned…", "I'll apply this by…". On this excerpt Example 1 would most likely route to **Tier-2 review** rather than straight to credit, which agrees with its low d6 score of 2.

SCOPE AND LIMITATIONS

These scores were produced by applying the production rubric and weights to the two transcripts as an expert reading. They are **not** a system of record: no CLE row, AI evaluation, or auditor score was created, and neither transcript passed through the live scoring pipeline.

A production run would additionally de-identify the transcript at the RIMPL boundary before any cloud call, apply the C1–C6 gate against real session metadata, and — under the blinded pilot configuration — withhold the AI score until an expert auditor has submitted theirs, so that the pair contributes to the ICC.

APPENDIX

Scored transcripts

Physician turns are marked; AI turns are shown for context only and earn no credit.

Example 1 — Acute watery diarrhoea cluster `48.0`

PHYSICIAN · T1

I'm covering a village health post and have seen more cases of acute watery diarrhea than usual over the past week, mostly in adults from a few neighboring households. What should I be considering?

AI · T1

A cluster of acute watery diarrhea across neighboring households raises concern for a common-source exposure... and warrants basic outbreak investigation — case counts by household and date of onset, a look at shared water or food sources, and consideration of stool testing for common pathogens if feasible. [Grade B]

PHYSICIAN · T2 `GENERIC — CAPS D1`

Is there a general approach for deciding whether this counts as an outbreak that needs to be reported?

AI · T2

Surveillance systems generally define an outbreak as an increase above what is normally expected for that place and time period, rather than a fixed case count... local or national reporting thresholds vary by jurisdiction. [Grade B]

PHYSICIAN · T3 `BEST TURN — D3/D4`

The households affected do share a shallow well. Does that change your thinking?

AI · T3

A shared shallow well is a plausible common source, particularly for waterborne pathogens, and would support prioritizing water source investigation and safe water messaging while confirmation is pending. [Grade B]

PHYSICIAN · T4 `ACCEPTANCE — D5 = 1`

Understood — I'll notify the district health office and advise the households on safe water practices in the meantime.

AI · T4

That is a reasonable interim step while investigation continues.

PHYSICIAN · T5 · CLOSING `ACTION, NOT LEARNING — D6 = 2`

Good to know. I'll keep monitoring case counts over the next few days.

Transcript reproduced as supplied; AI turns abridged where marked "..."

APPENDIX

Scored transcripts · II

Example 2 — Rising HIV test positivity 93.0

PHYSICIAN · T1
At the community clinic I support, the HIV test positivity rate has clearly risen over the past 6 months compared with the year before. What could explain this?

AI · T1
A rise in positivity could reflect a genuine increase in incidence, an increase in testing coverage or uptake among higher-risk individuals, improved test sensitivity, or some combination; a rise in detection alone cannot distinguish a true increase from a change in who is being tested. [Grade B]

PHYSICIAN · T2 `CHALLENGE + COMMIT — D5 = 5`
Are you just attributing this to testing bias? I think there may be a genuine increase in transmission in a specific risk group here.

AI · T2
I'm not asserting it is only a testing artifact — both explanations are plausible and need to be examined systematically. A useful first step is comparing the trend in testing volume against the trend in positivity rate... [Grade B]

PHYSICIAN · T3 `DEMANDS EVIDENCE GRADE — D1/D7`
Is there a standard epidemiological approach for telling apart a real increase in incidence from a testing artifact? I want an evidence grade on that, not just a general statement.

AI · T3
Practice generally recommends tracking testing volume alongside positivity rate over time and triangulating with sentinel or behavioral surveillance data... [Grade B] The specific statistical methods vary across surveillance systems, and I would not want to imply a single universally validated method exists. [?]

PHYSICIAN · T4 `DECISIVE CONTEXT — D3`
This area recently started outreach testing targeting migrant workers, and there have been informal reports of increased risk behavior locally. How does that factor in?

AI · T4
Both factors are plausible contributors... Stratifying the data by testing channel — outreach versus routine clinic testing — would help separate these two effects. [Grade B]

PHYSICIAN · T5 `OWNERSHIP + REVISION — D2/D7`
Then I'll request the positivity data stratified by testing channel before concluding this represents a real increase in transmission.

AI · T5
That is a reasonable next step; keep in mind that aggregate clinic-level data has inherent limitations even after stratification... [Grade B]

PHYSICIAN · T6 · CLOSING `TRANSFERABLE RULE — D6 = 5`
This case has changed how I'll approach similar surveillance signals going forward — I won't conclude that a rise in detection reflects a real increase in transmission until I've ruled out changes in testing coverage or case-finding strategy, especially when a new outreach program has started around the same time.